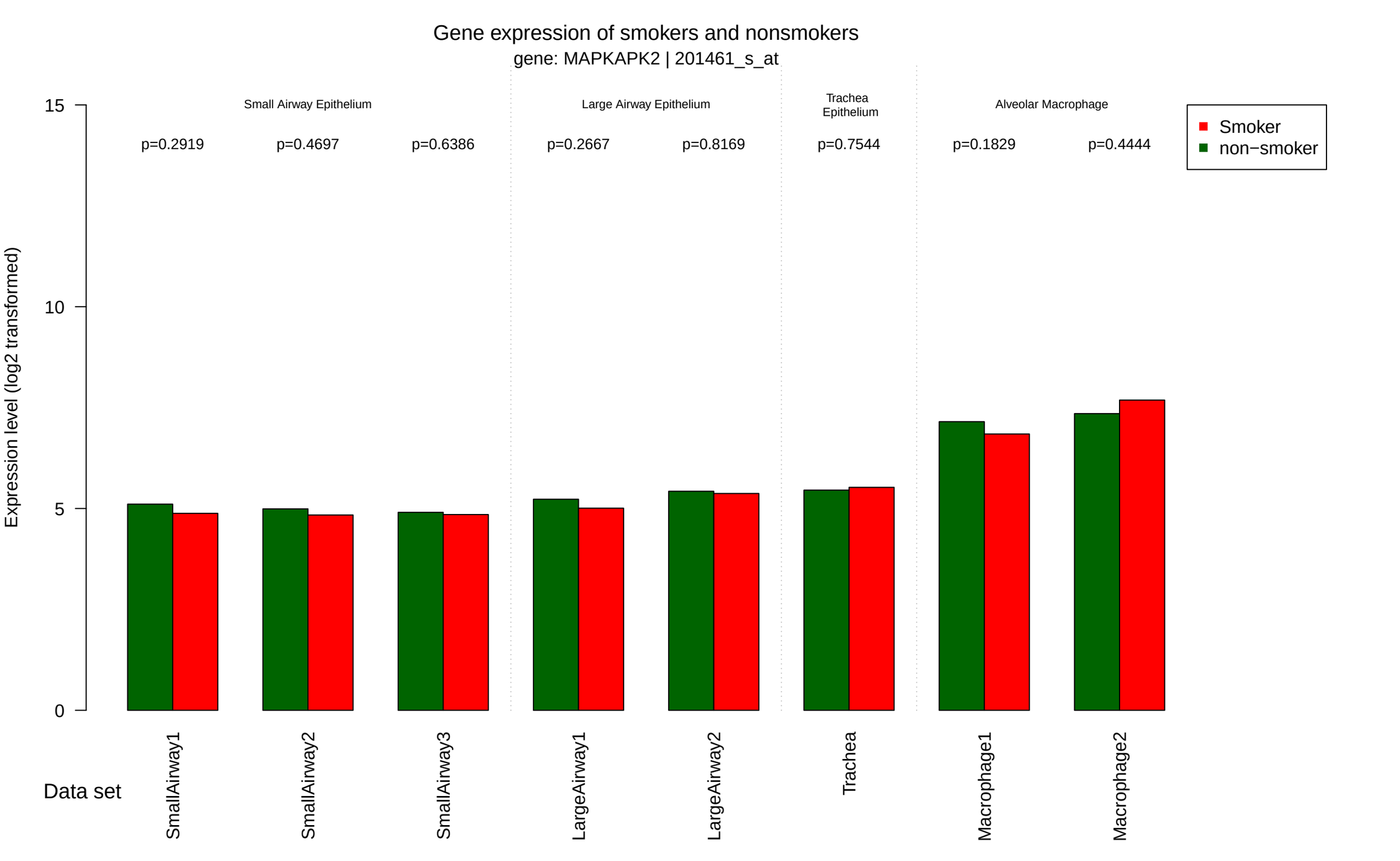
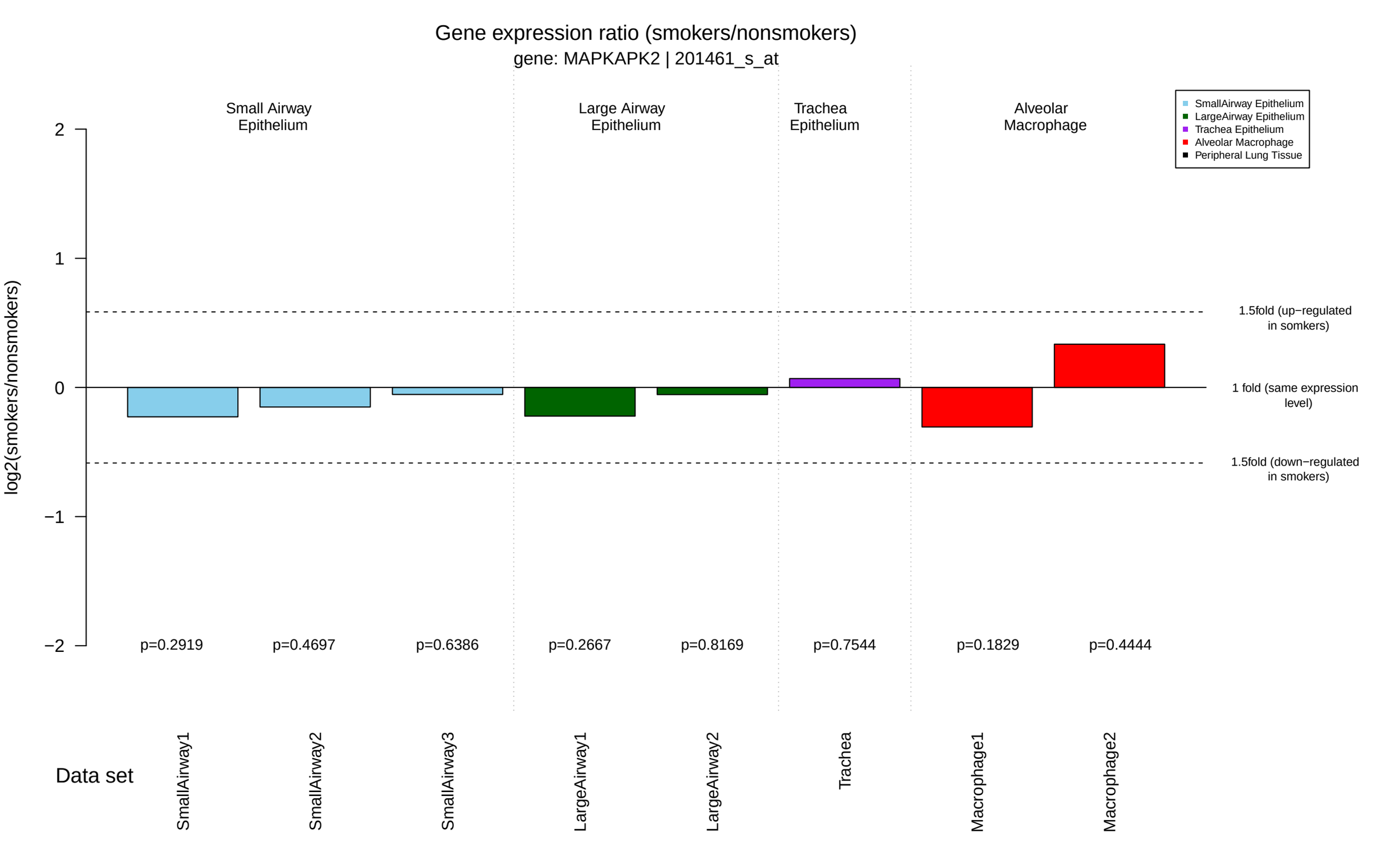
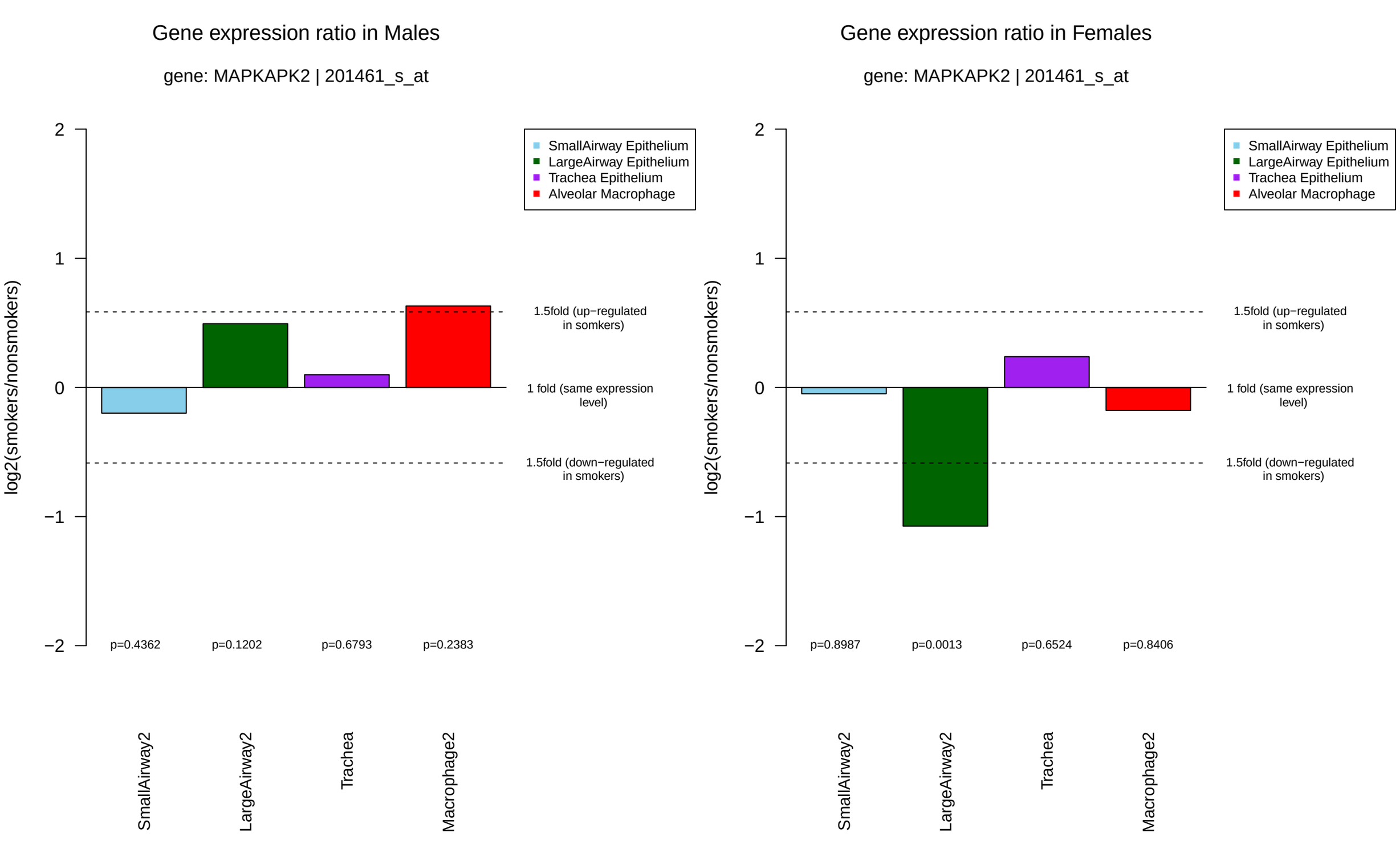
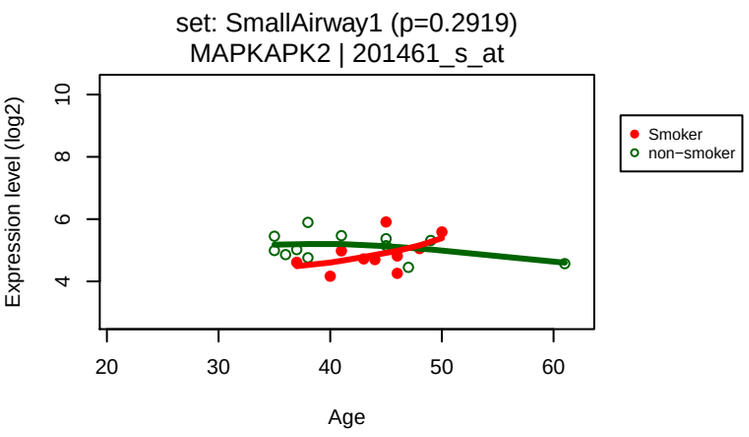
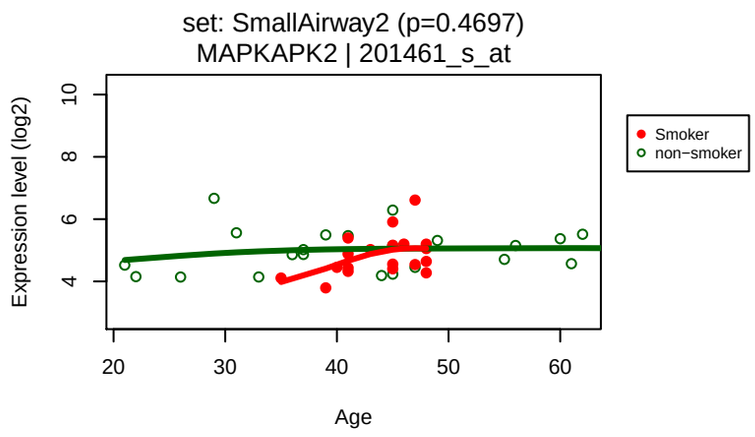
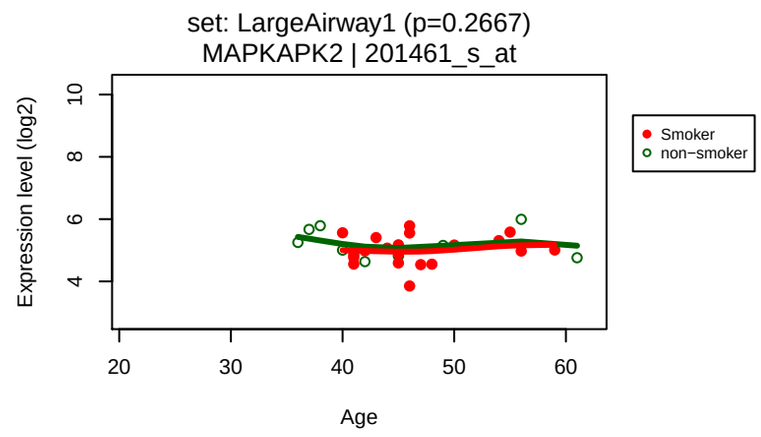

Gene expression of smokers and nonsmokers
gene: MAPKAPK2 | 201461_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2919
p=0.4697
p=0.6386
p=0.2667
p=0.8169
p=0.7544
p=0.1829
p=0.4444
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

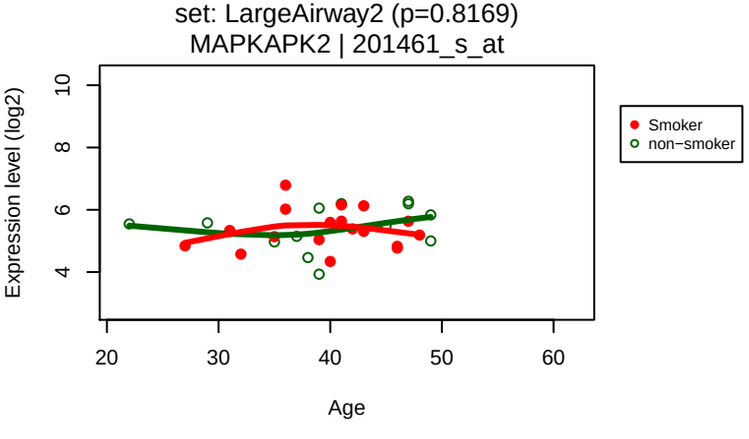

Gene expression ratio (smokers/nonsmokers)
gene: MAPKAPK2 | 201461_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2919
p=0.4697
p=0.6386
p=0.2667
p=0.8169
p=0.7544
p=0.1829
p=0.4444
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

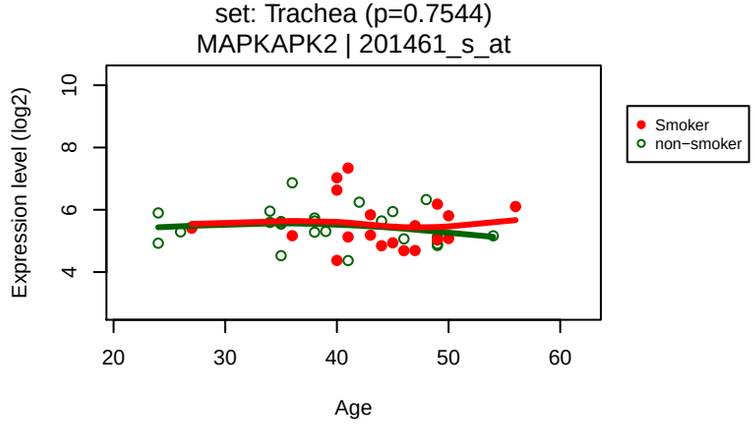

Gene expression ratio in Males
gene: MAPKAPK2 | 201461_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4362
p=0.1202
p=0.6793
p=0.2383
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPKAPK2 | 201461_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8987
p=0.0013
p=0.6524
p=0.8406
SmallAirway2
LargeAirway2
Trachea
Macrophage2

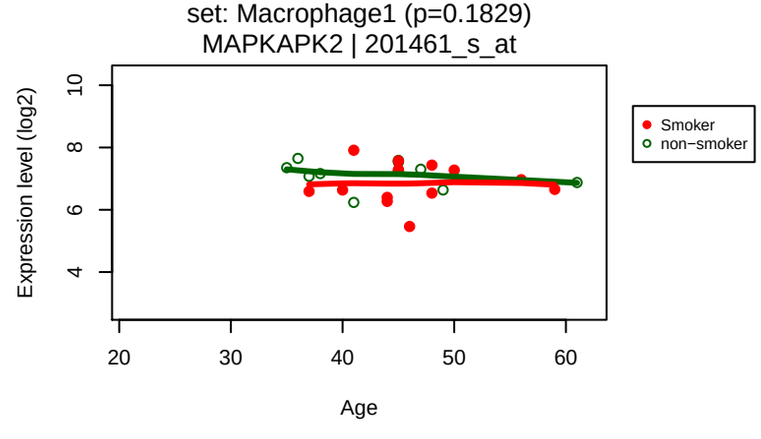

set: SmallAirway1 (p=0.2919)
MAPKAPK2 | 201461_s_at
Expression level (log2)
Age
Smoker
non−smoker

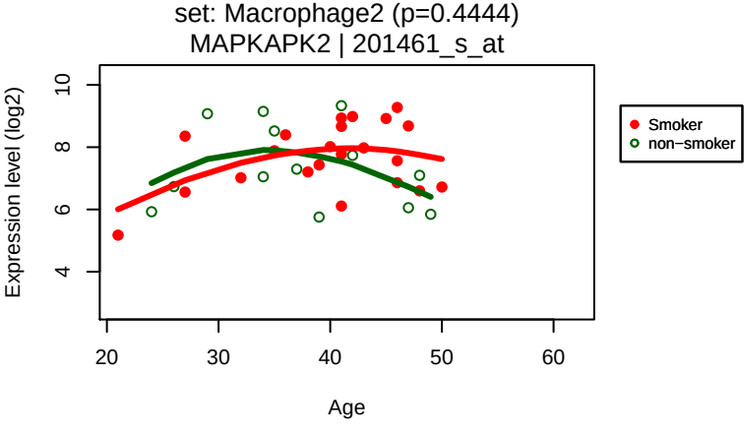

set: SmallAirway2 (p=0.4697)
MAPKAPK2 | 201461_s_at
Expression level (log2)
Age
Smoker
non−smoker

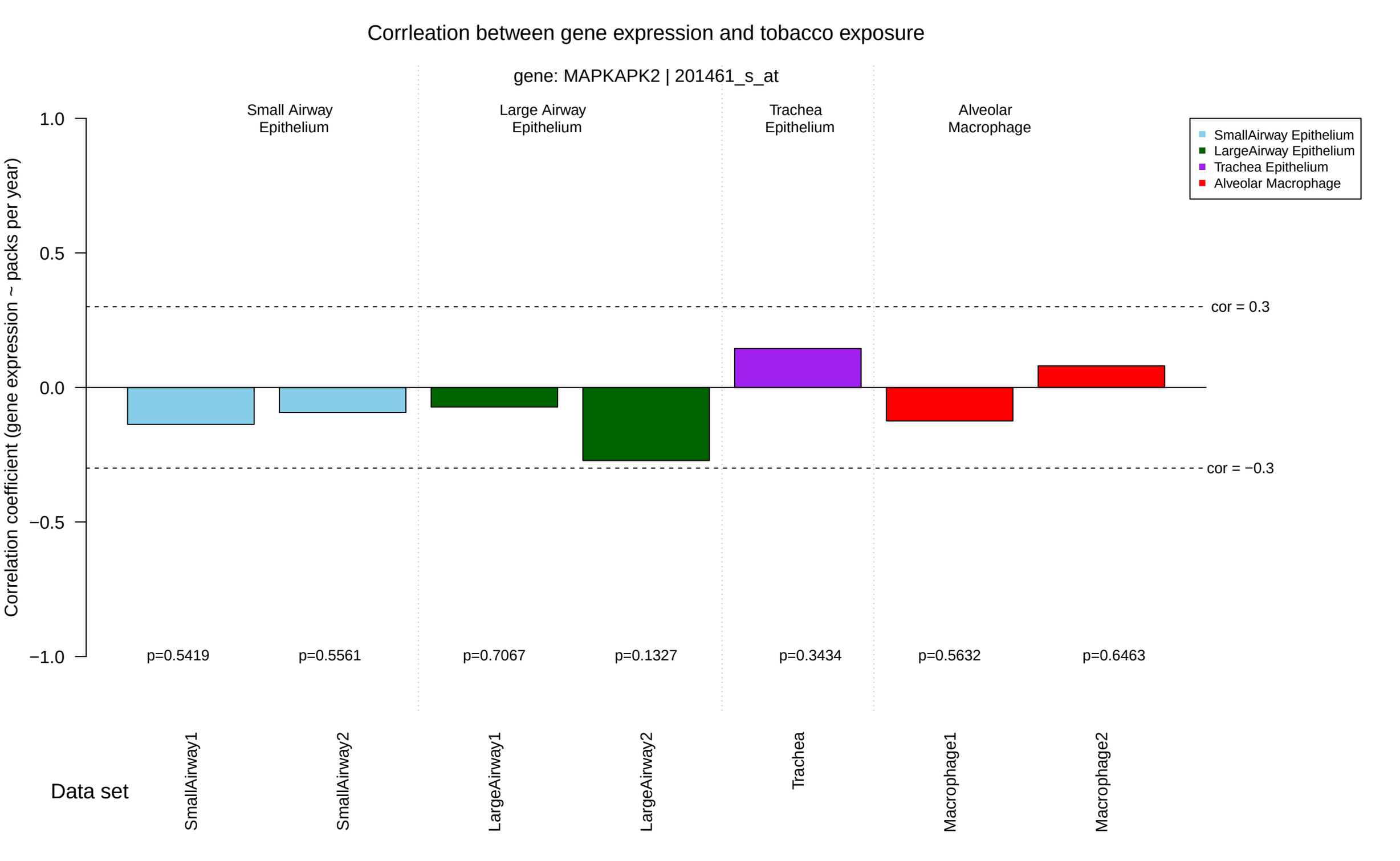

set: LargeAirway1 (p=0.2667)
MAPKAPK2 | 201461_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8169)
MAPKAPK2 | 201461_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7544)
MAPKAPK2 | 201461_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1829)
MAPKAPK2 | 201461_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4444)
MAPKAPK2 | 201461_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPKAPK2 | 201461_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5419
p=0.5561
p=0.7067
p=0.1327
p=0.3434
p=0.5632
p=0.6463
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2